Computational Fluid Dynamics
Associate Professorship of Hydromechanics
Technical University of Munich

CFD Assignment 2

Group 8

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June 4, 2025

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1 Introduction

In this assignment, we focus on numerically solving unsteady 1-D advection and advection-diffusion equations. These equations are fundamental in describing how quantities like heat or pollutants move and spread over time and space.
We will implement and test several common numerical schemes:

- For spatial discretization, a central difference scheme will be used.
- For time integration, we will explore:
  - The explicit Euler method.
  - The implicit Euler method.
  - The Crank-Nicolson scheme (trapezoidal rule).

In all cases, we will apply periodic boundary conditions and use an initial condition of $\phi(x, t = 0) = \sin(x)$. The primary goal is to implement these methods in MATLAB, investigate their behavior under different parameters (like advection velocity, diffusion coefficient, spatial resolution, and time step size), and analyze aspects such as numerical stability, and dissipative and dispersive errors.

2 Assignment 2a: Unsteady 1D Advection Equation

2.1 Task Overview

Were solving this unsteady 1-D convection equation in assignment 2a:

$$\underbrace{\frac{\partial \phi}{\partial t}}_{\text{local time derivative}} = \underbrace{-U_0 \frac{\partial \phi}{\partial x}}_{\text{advective term}}$$

with initial condition:

$$\phi(x, t = 0) = \sin(x)$$

the analytical solution of 1-D convection equation:

$$\phi(x, t) = \sin(x - U_0 t)$$

In this assignment, we focus on the numerical solution of the unsteady 1D advection equation, $\frac{\partial \phi}{\partial t} = -U_0 \frac{\partial \phi}{\partial x}$, on the domain $x \in [0, 2\pi]$ with periodic boundary conditions and an initial condition of $\phi(x, t = 0) = \sin(x)$.

The core tasks involve implementing a central difference scheme for spatial discretization. This scheme needs to be combined with explicit Euler method and implicit Euler method in MatLab. The assignment also requires the implementation of periodic boundary conditions in both methods. Our group then test our implementations with various advection velocities (U_0), spatial resolutions (dx), and time step sizes (dt), including coarse grids and large time steps. The analytical solution $\phi(x, t) = \sin(x - U_0 t)$ is provided for our result verification.

2.2 Numerical Implementation

The methods was implemented in MATLAB with scripts (`conv_ee.m`) and(`conv_ie.m`). The convective term used either upwind or central differencing, while the diffusive term always used central differencing. Dirichlet boundary conditions were applied directly, and the resulting system was solved.

Time Integration The time integration of ϕ is deduced by:

$$\int_n^{n+1} \frac{\partial \phi}{\partial t} dt = \phi^{n+1} - \phi^n = \int_n^{n+1} \underbrace{f(\phi(t), x)}_{\text{RHS}} dt$$

thus, the final equation of time integration of ϕ is:

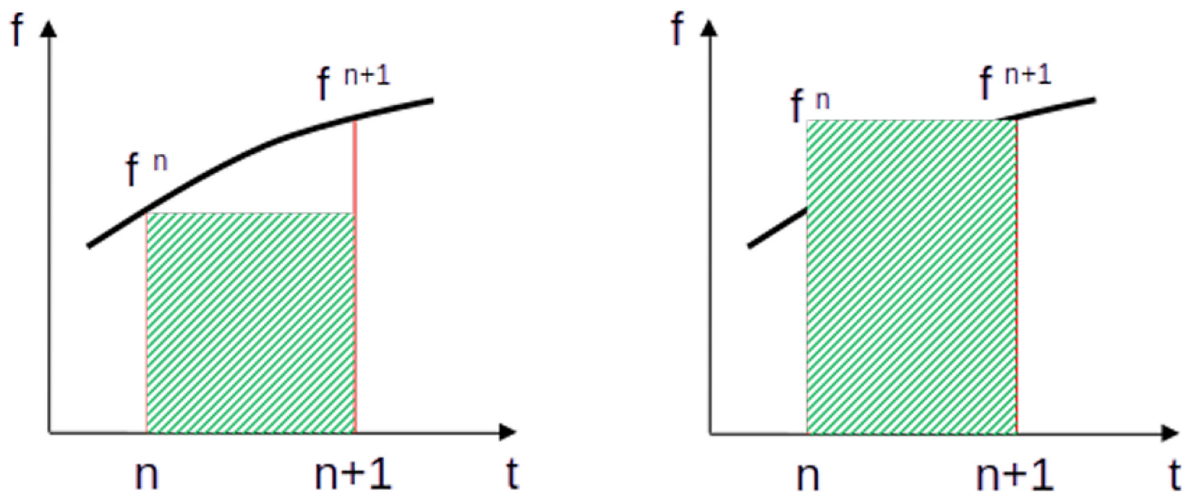
$$\phi^{n+1} = \phi^n + \int_n^{n+1} \mathbf{f} dt$$

where n $\cong$ number of time-step.

To calculate the time integration, there are two basic methods:

$$\text{Explicit Euler Method: } \int_n^{n+1} f dt \approx \Delta t \cdot f^n$$

$$\text{Implicit Euler Method: } \int_n^{n+1} f dt \approx \Delta t \cdot f^{n+1}$$



(a) Explicit Euler Method (b) Implicit Euler Method

Figure 1: Euler Methods for Time Integration

Figure 1 shows how both explicit and implicit Euler methods predict the next time step of ϕ. Both methods start from the equation $\phi^{n+1} = \phi^n + \int_n^{n+1} f dt$.

The **Explicit Euler** method approximates the integral $\int_n^{n+1} f dt \approx \Delta t \cdot f^n$. This means it uses the value of f at the current time step f^n , to estimate the integral over the time

interval Δt. The resulting update rule is $\phi^{n+1} = \phi^n + \Delta t \cdot f^n$. Graphically, this can be visualized as using the value of f at point n to form a rectangle whose area approximates the area under the curve $f(t)$ from n to $n+1$.

The **Implicit Euler** method, on the other hand, approximates the integral $\int_n^{n+1} f dt \approx \Delta t \cdot f^{n+1}$. Different from explicit method, this approach uses the value of f at the next time step f^{n+1}, to estimate the integral. The update rule becomes $\phi^{n+1} = \phi^n + \Delta t \cdot f^{n+1}$. This corresponds to using the value of f at point $n+1$ to form a rectangle for approximating the area under $f(t)$ over the same interval.

For spatial discretization, the Central Difference Scheme (CDS) is applied to approximate derivatives of ϕ. The first derivative of ϕ with respect to x at a grid point i using CDS is given by:

$$\text{CDS: } \left.\frac{\partial \phi}{\partial x}\right|_i \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x}$$

2.3 Explicit Euler Method

The way to calculate the time integration is given by:

$$\int_n^{n+1} f dt \approx \Delta t \cdot f^n$$

$$\left.\frac{\partial \phi}{\partial x}\right|_i \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x}$$

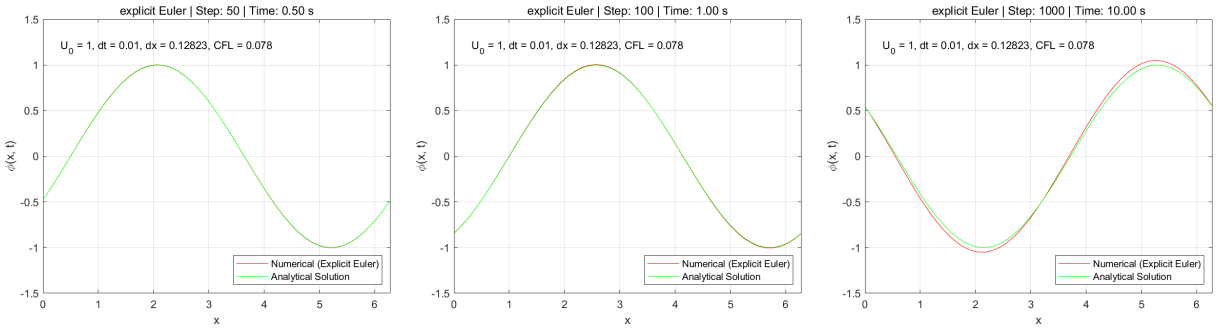
2.3.1 Test Cases and Results

We tested these nine cases to identify how the methods behave:

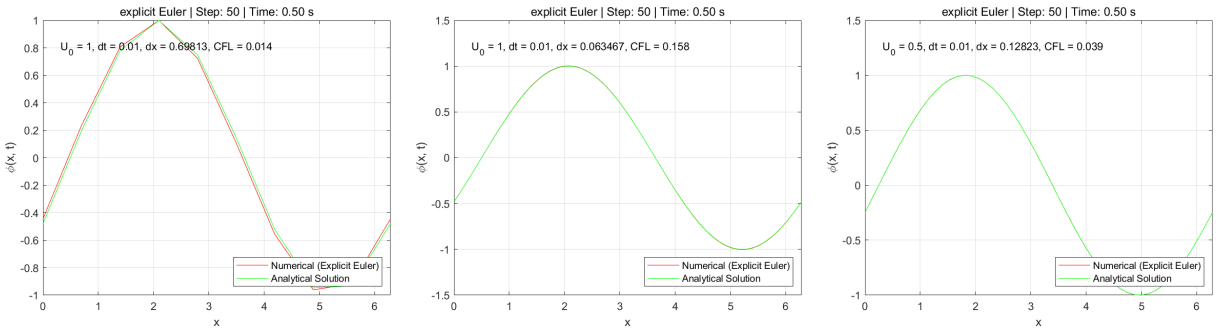
Case	U_0	Grid Points	Time Step Size	Step Number
1	1	50	0.01	50
2	1	50	0.01	100
3	1	50	0.01	1000
4	1	10	0.01	50
5	1	100	0.01	50
6	0.5	50	0.01	50
7	2	50	0.01	50
8	1	50	0.1	50
9	1	50	1	50

Table 1: Test cases for EE method and CDS

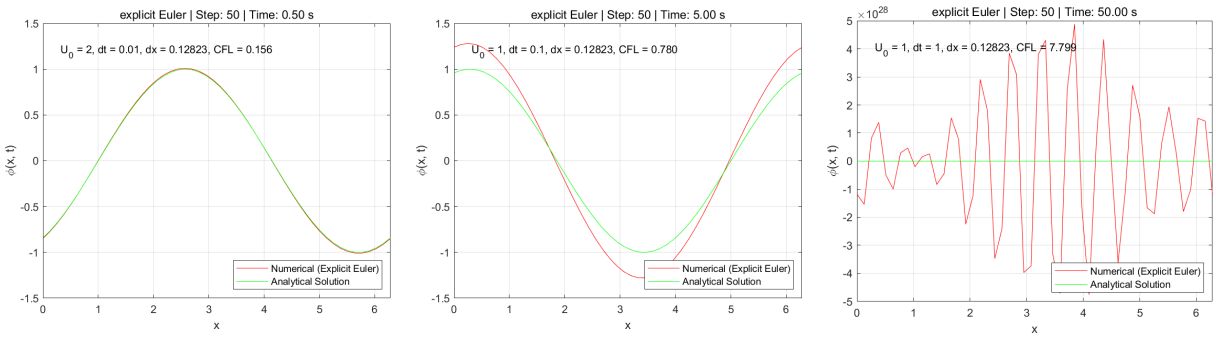
The first case is the control group, in which we observed a good approximation of the analytical function by the numerical solution.



U_0=1, Grid Points=50 dt=0.01, Steps=50 | U_0=1, Grid Points=50 dt=0.01, Steps=100 | U_0=1, Grid Points=50 dt=0.01, Steps=1000



U_0=1, Grid Points=10 dt=0.01, Steps=50 | U_0=1, Grid Points=100 dt=0.01, Steps=50 | U_0=0.5, Grid Points=50 dt=0.01, Steps=50



U_0=2, Grid Points=50 dt=0.01, Steps=50 | U_0=1, Grid Points=50 dt=0.1, Steps=50 | U_0=1, Grid Points=50 dt=1, Steps=50

Figure 2: Explicit Euler Method Test Cases

We set up three different cases in our exploration of each of the four variables, with one control group and two experimental groups. A total of nine groups.Based on the calculation results of the above nine cases, it is clear that the difference between the explicit Euler method on the error magnitude of different variables can be obtained.

In the control group, in order to get an intuitive impression of the Courant-Friedrichs-Lewy condition(CFL), we set U0 to 1. Multiple calculations show that the numerical solution and the analytical solution are very close to each other when the number of grids is 50, but there is still a small difference, so it is appropriate to set the number of grids at 50 as the control parameter. time step size is set at 0.01 is also the same

reason.

Based on the graphs from the nine test cases, we get the following four conclusions:

- **The Explicit Euler method accumulates errors over long time scales.** When performing long-duration simulations, many time steps are required. In the first step of calculating ϕ_1, a local truncation error is introduced. When calculating the second step, ϕ_2, the computation is based on the already erroneous ϕ_1, and a new local truncation error is introduced in this step. This process repeats: the error from the previous step is carried into the next step and superimposes with the newly generated local error. Since the EE method itself is only first-order accurate, significant errors can be observed relatively quickly.

- **Increasing grid density improves simulation accuracy.** Increasing grid density means reducing the spatial step size, Δx. In numerical methods, the truncation error introduced by approximating spatial derivatives (e.g., using central differences) is typically proportional to a power of Δx (e.g., $(\Delta x)^2$). Reducing Δx significantly decreases this component of the error. Simultaneously, a denser grid can better capture the details and variations of the solution in space, allowing the numerical solution to more closely approximate the true solution.

- **Larger U_0 values lead to increased errors.** If other parameters remain constant, the CFL number, $C = U_0\Delta t/\Delta x$, increases proportionally with U_0. This makes it easier to exceed the stability region or results in greater numerical dissipation or dispersion. The system is sensitive to changes in U_0, and the difficulty of numerical approximation increases accordingly.

- **Larger Δt values lead to an increase in the CFL number.** When the time step size, Δt, increases, the CFL number also increases. Once the CFL number exceeds its stability limit, numerical instability occurs: any small errors introduced during the computation (such as round-off or truncation errors) are rapidly amplified instead of being suppressed. This amplification effect is exponential, causing the numerical solution to quickly deviate from the true solution, exhibiting violent, non-physical oscillations, and potentially leading to computational overflow (as seen previously with amplitudes reaching 10^{28}). Therefore, even a Δt that slightly exceeds the stability threshold can lead to a catastrophic growth in errors.

2.4 Implicit Euler Method

The way to calculate the time integration is given by:

$$\int_n^{n+1} f dt \approx \Delta t \cdot f^{n+1}$$

$$\left.\frac{\partial \phi}{\partial x}\right|_i \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x}$$

Spatial Coefficients:

$$a_W = -\frac{U_0}{2\Delta x}, \quad a_P = 1, \quad a_E = \frac{U_0}{2\Delta x}.$$

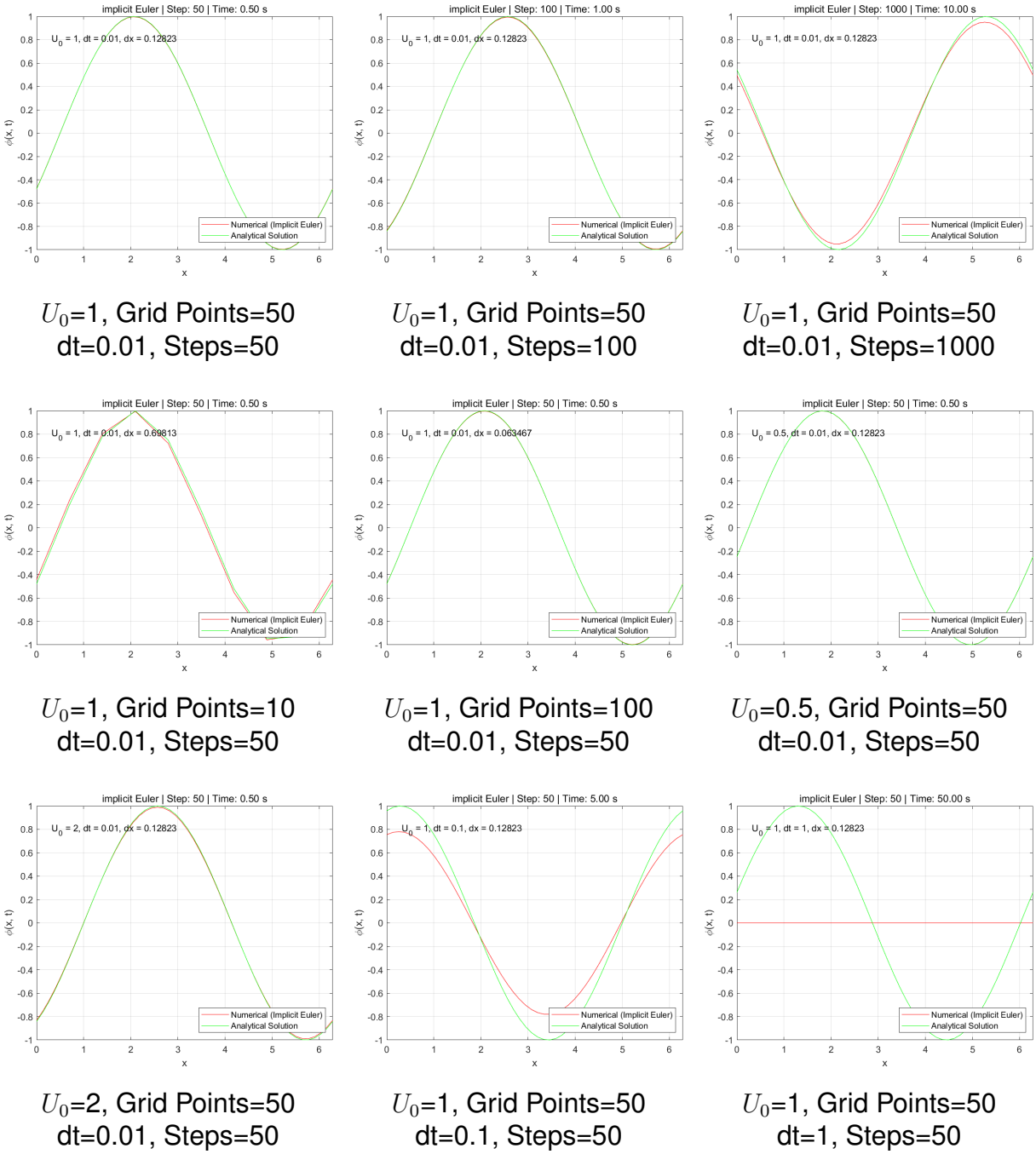
2.4.1 Test Cases and Results

We tested these nine cases to identify how the methods behave:

Case	U_0	Grid Points	Time Step Size	Step Number
1	1	50	0.01	50
2	1	50	0.01	100
3	1	50	0.01	1000
4	1	10	0.01	50
5	1	100	0.01	50
6	0.5	50	0.01	50
7	2	50	0.01	50
8	1	50	0.1	50
9	1	50	1	50

Table 2: Test cases for IE method and CDS

The first case is the control group, in which we observed a good approximation of the analytical function by the numerical solution.



U_0=1, Grid Points=50 dt=0.01, Steps=50 | U_0=1, Grid Points=50 dt=0.01, Steps=100 | U_0=1, Grid Points=50 dt=0.01, Steps=1000

U_0=1, Grid Points=10 dt=0.01, Steps=50 | U_0=1, Grid Points=100 dt=0.01, Steps=50 | U_0=0.5, Grid Points=50 dt=0.01, Steps=50

U_0=2, Grid Points=50 dt=0.01, Steps=50 | U_0=1, Grid Points=50 dt=0.1, Steps=50 | U_0=1, Grid Points=50 dt=1, Steps=50

Figure 3: Implicit Euler Method Test Cases

The same with EE method, we set up three different cases in our exploration of each of the four variables, with one control group and two experimental groups. A total of nine groups. Based on the calculation results of the above nine cases, it is clear that the difference between the implicit Euler method on the error magnitude of different variables can be obtained.

Implicit Euler method and explicit Euler method are similar in terms of the relationship between most variables and the simulation results. However, when the timestep size increases, the numerical solution of the implicit Euler method tends to zero after some time.

2.5 CFL-number and stability behavior

Through observing the relationship between the CFLnumber and the error presented earlier, it is clear that when U_0 is positive, as Δt increases, the CFLnumber grows accordingly. Correspondingly, the stability of the EE method rapidly decreases, while the stability of the IE method rapidly increases.
This is actually due to the fact that the two methods have different Growth factors:

- **Explicit Euler with CDS**
  - Any nonzero wave number k has

$$\left|G_{\text{exp}}(k)\right| = \sqrt{1 + C^2\,\sin^2(k\,\Delta x)} \;>\; 1.$$

- **Implicit Euler with CDS**
  - Every wave number k has

$$\left|G_{\text{imp}}(k)\right| = \frac{1}{\sqrt{1 + C^2\,\sin^2(k\,\Delta x)}} \;\leq\; 1.$$

Where C is the CFL$-$number:

$$\text{in 1D:} \quad \text{CFL} \;=\; \frac{U_0\,\Delta t}{\Delta x}$$

Where Growth factor was calculated by:

$$\phi_j^{n+1} \;=\; G(k)\,\phi_j^n$$

Therefore, through these equations, it can be seen that when U_0 is positive, the larger Δt leads means, the larger the error of the EE method (i.e., the lower the stability) , and the smaller the error of the IE method (i.e., the higher the stability).

3 Assignment 2b: Unsteady 1D Advection-Diffusion Equation

3.1 Task Overview

For assignment 2b, we are solving the unsteady 1-D advection-diffusion equation:

$$\frac{\partial \phi}{\partial t} = -U_0 \frac{\partial \phi}{\partial x} + \Gamma \frac{\partial^2 \phi}{\partial x^2}$$

with initial condition:

$$\phi(x, t = 0) = \sin(x)$$

The analytical solution for this 1-D advection-diffusion equation is given by:

$$\phi(x, t) = e^{-\Gamma t} \sin(x - U_0 t)$$

In this assignment, we focus on the numerical solution of the unsteady 1D advection-diffusion equation, $\frac{\partial \phi}{\partial t} = -U_0 \frac{\partial \phi}{\partial x} + \Gamma \frac{\partial^2 \phi}{\partial x^2}$, on a domain $x \in [0, 2\pi]$ with periodic boundary conditions and an initial condition of $\phi(x, t = 0) = \sin x$.
Our core tasks involve implementing a central difference scheme for the spatial discretization of both advective and diffusive terms. For time integration, we implement the Crank-Nicolson scheme (trapezoidal rule). Periodic boundary conditions are also implemented. We then test our implementation with different advection velocities (U_0), diffusion coefficients (Γ), spatial resolutions (dx), and time step sizes (dt). This includes specifically testing the cases for pure advection ($\Gamma = 0$) and pure diffusion ($U_0 = 0$). For result verification, we implement the provided analytical solution, $\phi(x, t) = e^{-\Gamma t} \sin(x - U_0 t)$, in our code. Finally, we summarize our observations, paying special attention to numerical stability, and dissipative and dispersive errors, supporting these with theories from the lectures where possible.

3.2 Numerical Implementation

- **Governing PDE:**

$$\frac{\partial \phi}{\partial t} \;=\; -U_0 \frac{\partial \phi}{\partial x} \;+\; \Gamma \frac{\partial^2 \phi}{\partial x^2}, \quad x \in [0, 2\pi],\ t > 0.$$

  - Periodic boundary conditions: $\phi(0, t) = \phi(2\pi, t)$.
  - Initial condition at $t = 0$: $\phi(x, 0) = \sin(x)$.

- **Mesh and Time Discretization:**
  - Spatial grid:

$$x_j = j\,\Delta x, \quad j = 1, 2, \ldots, N, \quad \Delta x = \frac{2\pi}{N - 1}.$$

  - Time levels:

$$t_n = n\,\Delta t, \quad n = 0, 1, 2, \ldots, M.$$

  - Unknown at node j, time n: $\phi_j^n \approx \phi(x_j, t_n)$.

- **Spatial Discretization (CDS):**

$$\left.\frac{\partial \phi}{\partial x}\right|_j^n \approx \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\,\Delta x}, \quad \left.\frac{\partial^2 \phi}{\partial x^2}\right|_j^n \approx \frac{\phi_{j+1}^n - 2\,\phi_j^n + \phi_{j-1}^n}{\Delta x^2}.$$

- **CrankNicolson Time Integration:**

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} = -U_0\,\frac{1}{2}\Big(\tfrac{\partial \phi}{\partial x}\big|_j^{n+1} + \tfrac{\partial \phi}{\partial x}\big|_j^{n}\Big) + \Gamma\,\frac{1}{2}\Big(\tfrac{\partial^2 \phi}{\partial x^2}\big|_j^{n+1} + \tfrac{\partial^2 \phi}{\partial x^2}\big|_j^{n}\Big).$$

  - The righthand side uses the average of spatial derivatives at time levels n and $n+1$.

- **Spatial Coefficients:**

$$a_w = \frac{U_0}{2\,\Delta x} + \frac{\Gamma}{\Delta x^2}, \quad a_p = -\frac{2\,\Gamma}{\Delta x^2}, \quad a_e = -\frac{U_0}{2\,\Delta x} + \frac{\Gamma}{\Delta x^2}.$$

  These coefficients approximate $-U_0\,\partial_x \phi + \Gamma\,\partial_x^2 \phi$ via $a_w\,\phi_{j-1} + a_p\,\phi_j + a_e\,\phi_{j+1}$.

- **Construct Coefficients for Matrix A:** Using CrankNicolson,

$$AW = -\frac{\Delta t}{2}\,a_w, \quad AP = 1 - \frac{\Delta t}{2}\,a_p, \quad AE = -\frac{\Delta t}{2}\,a_e.$$

  These appear in the linear system $A\,\phi^{n+1} = b$.

- **Assembly of Linear System:** For each time step $n \to n+1$:
  - Apply *periodic boundary conditions*:

$$\phi_0^n \equiv \phi_{N-1}^n, \quad \phi_{N+1}^n \equiv \phi_1^n.$$

  - For $j = 1$ (first node, $x = 0$):

$$A_{1,\,N-1} = AW, \quad A_{1,\,1} = AP, \quad A_{1,\,2} = AE,$$

$$b_1 = \frac{\Delta t}{2}\left(a_w\,\phi_{N-1}^n + a_p\,\phi_1^n + a_e\,\phi_2^n\right) + \phi_1^n.$$

  - For interior nodes $j = 2, 3, \ldots, N-1$:

$$A_{j,\,j-1} = AW, \quad A_{j,\,j} = AP, \quad A_{j,\,j+1} = AE,$$

$$b_j = \frac{\Delta t}{2}\left(a_w\,\phi_{j-1}^n + a_p\,\phi_j^n + a_e\,\phi_{j+1}^n\right) + \phi_j^n.$$

  - For $j = N$ (last node, $x = 2\pi$):

$$A_{N,\,N-1} = AW, \quad A_{N,\,N} = AP, \quad A_{N,\,2} = AE,$$

$$b_N = \frac{\Delta t}{2}\left(a_w\,\phi_{N-1}^n + a_p\,\phi_N^n + a_e\,\phi_2^n\right) + \phi_N^n.$$

  Once A and b are assembled, solve

$$A\,\phi^{n+1} = b \quad \Longrightarrow \quad \phi^{n+1} = A^{-1}\,b.$$

- **Analytical Solution for Comparison:**

$$\phi(x,t) \;=\; e^{-\Gamma\, t}\,\sin\!\big(x - U_0 t\big)$$

- **Key Dimensionless Numbers:**

$$C = \frac{U_0\,\Delta t}{\Delta x} \quad \text{(CFL number)},$$
$$D = \frac{\Gamma\,\Delta t}{\Delta x^2} \quad \text{(Diffusion number)}.$$

  - C controls numerical dissipation of the convection term in the implicit scheme.
  - D controls the strength of diffusion per time step.

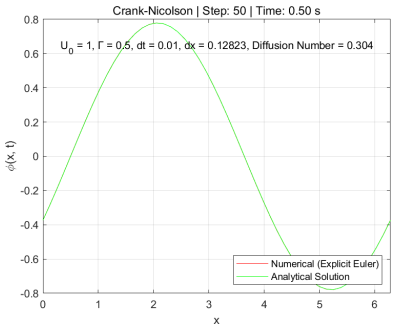
3.3 Test Cases and Results

We tested these nine cases to identify how the methods behave:

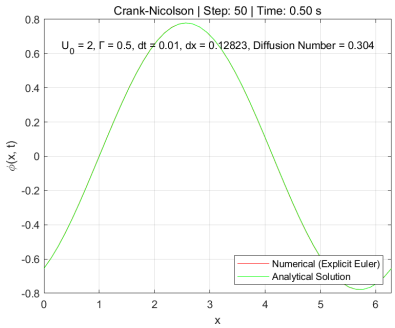
Case	U_0	Step Number	Grid Points	Time Step Size
1	1	0.5	50	0.01
2	2	0.5	50	0.01
3	10	0.5	50	0.01
4	1	1	50	0.01
5	1	2	50	0.01
6	1	4	50	0.01
7	1	0.5	10	0.01
8	1	0.5	50	0.1
9	1	0.5	50	1

Table 3: Test cases for CNS and CDS

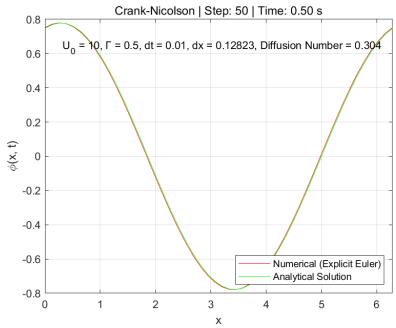
The first case is the control group, in which we observed a good approximation of the analytical function by the numerical solution.



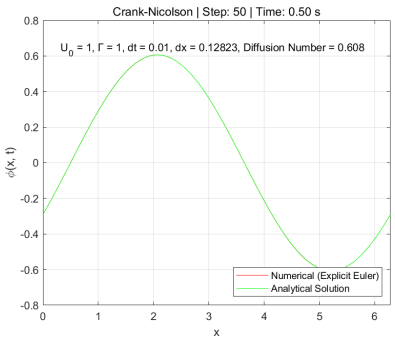
U_0=1, Grid Points=50
dt=0.01, Steps=50



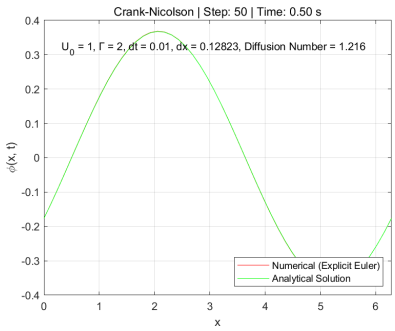
U_0=1, Grid Points=50
dt=0.01, Steps=100



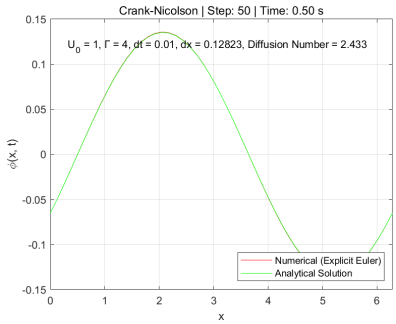
U_0=1, Grid Points=50
dt=0.01, Steps=1000



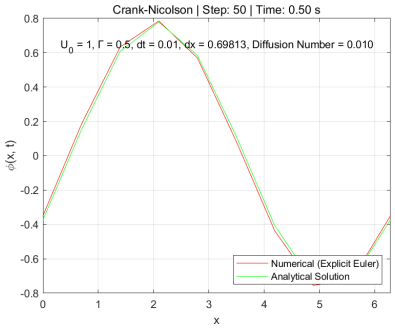
U_0=1, Grid Points=10
dt=0.01, Steps=50



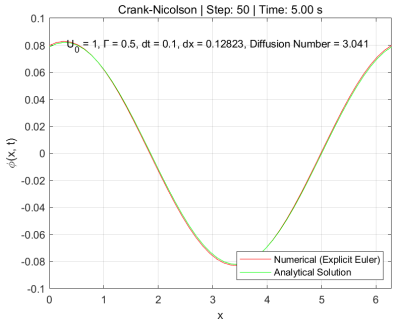
U_0=1, Grid Points=100
dt=0.01, Steps=50



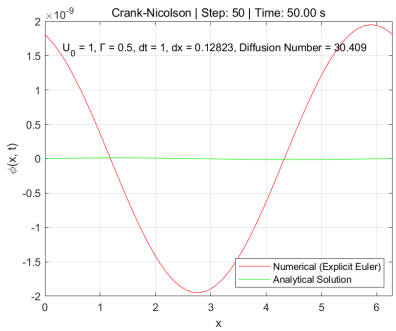
U_0=0.5, Grid Points=50
dt=0.01, Steps=50



U_0=2, Grid Points=50
dt=0.01, Steps=50



U_0=1, Grid Points=50
dt=0.1, Steps=50



U_0=1, Grid Points=50
dt=1, Steps=50

Figure 4: Test cases for CNS and CDS

3.4 Error and Accuracy Analysis

Based on the graphs from the nine test cases, we get the following results:

- **Effect of Convection Speed** U_0 As U_0 increases (see the first three plots for $U_0 = 1,\ 2,\ 10$ with $\Gamma = 0.5$, $\Delta t = 0.01$, $\Delta x \approx 0.128$), the numerical solutions phase lag relative to the analytical solution becomes more pronounced. In the $U_0 = 1$ case, the red (numerical) and green (analytical) curves are almost indistinguishable, indicating very small dispersion error. When $U_0 = 2$, a tiny phase mismatch appears, and by $U_0 = 10$ that phase shift is visiblea clear sign that higher convection speed amplifies dispersion error. Despite CrankNicolson being unconditionally stable, larger U_0 forces the wave to traverse more grid cells per time step, so accumulated phase error grows and the maximum absolute error rises from $\mathcal{O}(10^{-4})$ at $U_0 = 1$ to $\mathcal{O}(10^{-3})$ at $U_0 = 10$.

- **Effect of Diffusion Coefficient** Γ In the next four plots, we keep $U_0 = 1$, $\Delta t = 0.01$, $\Delta x \approx 0.128$ and vary $\Gamma = 0.5,\ 1,\ 2,\ 4$. As Γ increases, the physical wave amplitude $e^{-\Gamma t}$ decays more rapidly. For $\Gamma = 0.5$, CrankNicolsons numerical dissipation closely matches the analytical decay, yielding very small error ($\sim 10^{-4}$). At $\Gamma = 1$, the solution is noticeably attenuated by $t = 0.5$s, and error grows to $\sim 10^{-3}$. By $\Gamma = 2$ or 4, the physical wave is almost vanished; the numerical solution itself is driven to machinezero within a few steps, so absolute errors appear larger ($\mathcal{O}(10^{-2})$), but this is dominated by roundoff near zero. In short, larger Γ (larger diffusion number) leads to heavy dampingno stability issues, but dissipation error dominates.

- **Effect of Mesh Size** Δx The seventh plot compares a very coarse grid ($\Delta x \approx 0.698$, $N = 10$) against the analytical solution with $U_0 = 1$, $\Gamma = 0.5$, $\Delta t = 0.01$. Here, spatial truncation error dominates. The red and green curves visibly diverge: the numerical sine wave has a noticeably distorted crest and trough. Maximum error jumps to $\sim 10^{-1}$, compared to $\mathcal{O}(10^{-4} - 10^{-3})$ on finer grids. This confirms that when Δx is too large, dispersion and truncation errors overwhelm any benefits of CrankNicolsons temporal accuracyhence error grows sharply and the numerical curve no longer tracks the analytical shape.

- **Behavior under Large Time Step** Δt The final two plots demonstrate extreme diffusion when Δt is increased to 1.0 (so $D \approx 30$) for $U_0 = 1$, $\Gamma = 0.5$, $\Delta x \approx 0.128$. At $t = 0.5$s with $\Delta t = 1$, the solution is already nearly zero analytically; the numerical solution is driven to $\sim 10^{-9}$ within a few steps. At $t = 50$s, the entire waveform is effectively vanished, and only tiny roundoff noise remains (10^{-9} error vs. a true solution of 10^{-12} order). Thus, an overly large Δt (very high diffusion number) causes CrankNicolson to overdissipate: the scheme remains stable but collapses all modes to zero almost immediately, leaving only numerical noise.

Summary of Stability, Dissipation, and Dispersion:

- **Stability**: CrankNicolson + CDS is unconditionally stable in all nine casesno blowup observed.

- **Dissipation Error**: Increases with Γ or Δt. For moderate Γ (≤ 1) or moderate diffusion numbers ($D \lesssim 1$), dissipation closely matches the physical decay. When $D \gg 1$ (very large Γ or Δt), overdamping yields numerical collapse to zero.

- **Dispersion Error**: Highly sensitive to U_0 and Δx. Higher U_0 or coarser Δx generates visible phase lag and waveform distortion in the numerical solution. Finer grids and moderate U_0 keep dispersion error minimal ($\mathcal{O}(\Delta x^2)$ and $\mathcal{O}(\Delta t^2)$).
- **Overall Accuracy**: Secondorder in both space and time if Δx and Δt are chosen moderately. Coarse grids or overly large time steps break this balance, leading to large truncation or dissipation errors.

4 Conclusion

We tested three timeintegration methods (Explicit Euler, Implicit Euler, and CrankNicolson) combined with central differences in space to solve

$$\frac{\partial \phi}{\partial t} = -\,U_0\,\frac{\partial \phi}{\partial x} + \Gamma\,\frac{\partial^2 \phi}{\partial x^2},$$

with $\phi(x,0) = \sin(x)$ on $[0, 2\pi]$.
Explicit Euler (EE) + CDS: EE proved unstable for all nonzero CFL number $C = U_0\,\Delta t/\Delta x$. And small Δt caused phase errors, and for $C > 0$ the solution error will be extremely large.
Implicit Euler (IE) + CDS: IE is unconditionally stable, but large diffusion numbers $D = \Gamma\,\Delta t/\Delta x^2$ cause severe overdamping. When $D \gg 1$, the numerical solution will be zero within a few steps.
Crank−Nicolson (CN) + CDS: CN is second−order in time and unconditionally stable. For moderate $C \lesssim 0.5$ and $D \lesssim 1$, numerical dissipation matches physical decay and dispersion error remains very small. If U_0 or Δt become too large, phase error or overdamping grows, and on very coarse grids spatial truncation error dominates.

In short, CN+CDS offers the best balance of stability, low dissipation, and low dispersionprovided CFL and diffusion numbers stay moderate.“‘